Module 7: Lesson 2

Reinforcement Learning: Temporal Difference Methods

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Outline

- Temporal Difference methods.
- SARSA
- Q-learning
- Application: The windy gridworld

Temporal Difference methods

Temporal-Difference (TD) learning combines the Monte Carlo simulation method and the Dynamic Programming (DP) notions of optimization.

As in Monte Carlo methods, TD methods learn directly from experience without a model of the environment.

As in DP, TD methods update estimates from states already updated, without waiting for a final outcome of an episode. This resembles the asynchronous dynamic programming techniques that we described in Module 2 of this course.

Some applications have very long episodes, so delaying all learning until the end of the episode slows down the optimization.

We consider two applications of Temporal Difference: SARSA and Q-Learning.

SARSA

The SARSA method considers transitions from state–action pairs to state–action pairs.

The estimation starts with a current observation of the state-action (s_t, a_t), which yields a reward $r(s_t, a_t)$ and transition to state-action (s_{t+1}, a_{t+1}).

"On-policy" algorithm: the actions a_t and a_{t+1} arise from the optimal actions given states s_t and s_{t+1}, respectively, using the current guess of policy, Π.

Given a vector of observations $(s_t, a_t, r_t, s_{t+1}, a_{t+1})$, which gives the name to the optimization method, we can update the state-action value as:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \big[r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t) \big]$$

where the parameter $\alpha \in (0, 1)$ captures the adjustment of our guesses to new information.

Q-Learning

Q-learning is an "off-policy" method that is defined by the following updating criterion:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)\right]$$

Instead of choosing the action a_{t+1} from the current policy (on-policy), we directly choose the action that maximizes the state-action value for the agent in step $t + 1$ given the observed transition to state s_{t+1} (off-policy).

In spite of that, the current policy still determines which actions are followed and updated in the next time step.

SARSA and Q-Learning in the windy gridworld

	Right	Right	Down
Up	Right	Right	Down
Right	Right	Right	Down
Right	Right	Right	

	Left	Right	Down
Up	Right	Right	Down
Right	Right	Down	Down
Right	Right	Right	

Both methods find similar choices. But Q-Learning seems to find the most consistent policy in cell 1. In contrast, SARSA prescribes a "right" movement in that cell, which is one step away, let wind not happen, from the goal.

This effect arises because the SARSA method learns from the ε-greedy choices that were successful later in reaching the bottom terminal cell.

Again, if we were to allow more exploration or more experience for the agent with more episodes, we should reach the trivially optimal "left" action.

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Summary of Lesson 2

In Lesson 2, we have looked at:

- The main insights of Temporal Difference methods
- SARSA and Q-Learning algorithms
- Application to the windy gridworld

⇒ **References for this lesson:**

Sutton, Richard S., and Andrew G. Barto. *Reinforcement Learning: An Introduction.* MIT Press, 2018. (see Chapter 6)

TO DO NEXT: Now, please go to the associated Jupyter Notebook for this lesson to study the implementation of SARSA and Q-Learning to the windy gridworld.

In the next lesson, we will analyze another implementation of TD to compare SARSA and Q-Learning.